

CLASS TEST, 2023

Data Structure and Algorithms (IT-2101)

Full Marks: 20

Time : 45 Minutes

Answer all the questions

1. Define binary tree and complete binary tree. 2
2. Given a Singly Linked List, write down a function that starting from the second node prints all alternate nodes of it. For example, if the given linked list is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$ then your function should print 1 3 5, and if the given linked list is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ then it prints 1 3. 3
3. Suppose $L1$ and $L2$ are two sorted single linked lists. Write down a procedure to merge them together to create another single sorted linked list $L3$. 4
4. Write algorithms to delete a node at k^{th} position in a doubly linked list. 2
5. How many pointers need to be changed during insertion in a doubly linked list? 2
6. Create a height balanced (AVL) tree using the following integer in a sequence: 55, 14, 10, 5, 25, 55, 30, 3, 33, 25. 4
7. Write an nonrecursive algorithm for preorder traversal of a tree. 3

3. A priority queue is an abstract data type which is like a regular queue data structure, but where additionally each element has a "priority" associated with it. In a priority queue, an element with high priority (lower value) is served before an element with low priority (higher value). Now consider the following variation of the priority queue. Let the range of the priorities be $[0; 50)$. Divide the priorities into 5 groups, $[0; 10)$, $[10; 20)$, $[20; 30)$, $[30; 40)$, and $[40; 50)$. An element with priority "a" is placed in group $[x; y)$ provided $x \leq a < y$. If two elements are in the same priority group, their values are compared and the lowest valued element is processed. Write a C program that will take a file input which will consist of 3 operations : Insert k, Delete and Display and perform the operations as explained above.

Input format:

i x y : i stands for insert; x stands for element; y stands for priority

p: p stands for display the queue

d: d stands for delete an element from the queue

Indian Institute of Engineering Science and Technology, Shibpur Date 24/11/2023
 B. Tech (Information Technology) 3rd Semester Class Test on Signals and Systems IT 2104
 Answer both the questions.

Full Marks: 10

Time- 30 Minutes

1.a) Using only the time shifting property, find the Fourier transform of the signal in Fig.1
 $g(t) = \sin t$

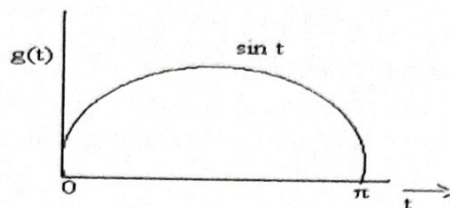


Fig. 1

b) Find the time auto-correlation function of the signal $g(t) = e^{-at} u(t)$ and from it determine ESD (energy spectral density) of $g(t)$.

2+3

2. a) If $X(t)$ and $Y(t)$ are both independent binary signals each taking on values -1 and 1 only with $P_X(1) = Q = 1 - P_X(-1)$ and $P_Y(1) = P = 1 - P_Y(-1)$
 Determine $P_Z(z_i)$ where $Z = X + Y$

b) Prove that the variance of a sum of independent RVs is equal to the sum of their variances. Also write the extension of this result for any number of RVs.

2+3

Indian Institute of Engineering Science and Technology, Shibpur
B. Tech (IT) 3rd Semester Final Examination, November 2023
Signals and Systems (IT 2104)

Answer any five questions

Full Marks: 50

Time- 3 hrs

1. a) Find E_x and E_y , the energies of the signals $x(t)$ and $y(t)$ shown in Fig. 1. Sketch the signals $x(t)+y(t)$ and $x(t) - y(t)$, and show that the energies of either of these two signals are equal to E_x+E_y . Are the two signals $x(t)$ and $y(t)$ orthogonal or non-orthogonal? Justify mathematically.

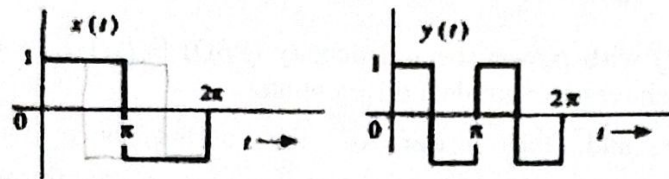


Fig.1

- b) For the signal $y(t)$ shown in Fig. 2, sketch the signals: (a) $g(t-4)$; (b) $g(t/1.5)$; (c) $g(2t-4)$; (d) $g(t+2)$; (e) $g(2-t)$

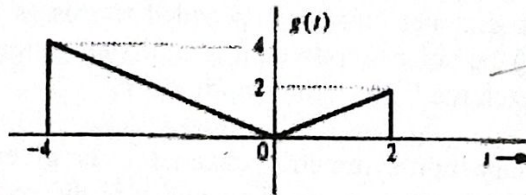


Fig. 2

(2+2+1)+5

- 2.a) Find the compact trigonometric Fourier Series for the periodic square wave $w(t)$ shown below in Fig. 3. Sketch its amplitude and phase spectra. If either the sine or the cosine terms are absent in the Fourier series, explain why.

$w(t)$

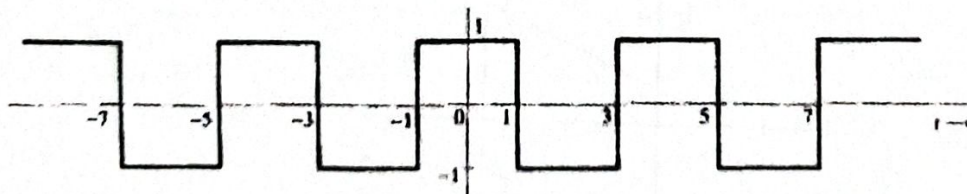


Fig. 3

- b) Derive the relation of autocorrelation function with energy spectral density (ESD) of an energy signal $g(t)$. Find the time autocorrelation function of the signal $g(t)=e^{-at} u(t)$ ($a>0$) and its ESD.

5+(2+3)

3. a) Verify the linearity/non-linearity and time variance/invariance of the systems
 i) $y(t)=2x(t-5)$; ii) $y(t)=2x(3t)$
 Where, signals $y(t)$ and $x(t)$ represent the system output and input signals, respectively.

- b) Find the Fourier transform of the function $g(t)=\text{Sgn } t$.
 c) A random binary pulse train has pulse width $T_b/2$ and one binary digit is transmitted every T_b second. A binary 1 is transmitted by the positive pulse and a binary 0 is transmitted by the negative pulse. The two symbols are equally likely and occur randomly. Determine the autocorrelation function, the PSD (power spectral density) and the essential bandwidth of the signal.

3+2+5

4. (a) Determine the sequence $g(n)$ which is obtained through linear convolution of the sequences $x(n)$ and $y(n)$ as follows:

$$x(n)=\{1, 2, 3, 1\} \text{ and } y(n)=\{1, 2, 1, -1\}$$

$\uparrow$ $\uparrow$
 $n=0$ $n=0$

- b) A noise signal $n_i(t)$ with power spectral density (PSD) $S_{ni}(w)=K$ is applied at the input to a system that behaves like an ideal differentiator.

Determine the PSD and the power of the output noise signal $n_o(t)$.

- c) Determine the maximum bandwidth of a signal when transmitted through the system that behaves like low-pass RC filter with $R=10^3 \Omega$ and $C=10^{-9} \text{ F}$ if, over this bandwidth, the amplitude response (gain) variation is to be within 5% and the time delay variation is to be within 2%.

3+2+5

5. (a) The random binary signal $X(t)$ can take on only two values, 3 and 0, with **equal** probability. A Gaussian channel noise $n(t)$ is added to this signal, giving the received signal $Y(t)$. The PDF of the noise amplitude n is Gaussian with a zero mean and an **rms** value of 2. Determine and sketch the PDF of the amplitude Y .

- b) For a certain binary non-symmetric channel it is given that $P_{y/x}(0/1)=0.1$ and $P_{y/x}(1/0)=0.2$, where x is the transmitted digit and y is the received digit. If $P_x(0)=0.4$, determine $P_y(0)$ and $P_y(1)$.

- c) The transmission power profile is an RV X that follows a PDF, as shown, in Fig.4 below. Find the mean, the mean square and the variance of X

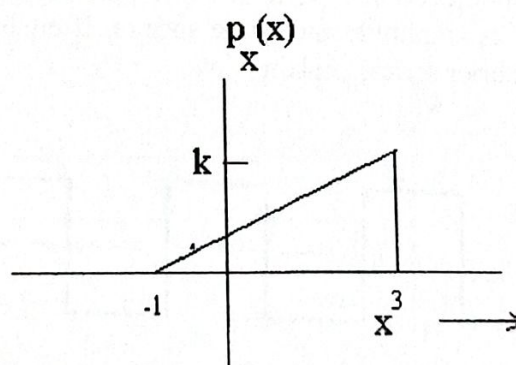


Fig. 4

4+3+3

6(a) Show that if two random variables x and y are related by $y=k_1 x+k_2$, where, k_1 and k_2 are arbitrary constants, the correlation coefficient $\rho_{xy}= 1$, if k_1 is positive and $\rho_{xy}= -1$, if k_1 is negative.

b) Over a certain binary communication channel, the symbol 0 is transmitted with probability 0.4, and 1 is transmitted with probability 0.6. It is given that $P(e/0)=10^{-6}$ and $P(e/1)=10^{-4}$, where $p(e/x_i)$ is the probability of detecting the error given that x_i is transmitted. Determine $P(e)$, the error probability of the channel.

c) The sum of the points on two dice thrown is a discrete random variable X . Find $\bar{X}$, $\bar{X}^2$ and σ_x^2 .

3+3+ 4

7. a) i) If $z=x+y$, where x and y are independent Gaussian RVs with

$$p_x(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\bar{x})^2/2\sigma_x^2} \text{ and } p_y(y) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(y-\bar{y})^2/2\sigma_y^2}$$

then show that z is also Gaussian with $\bar{z} = \bar{x} + \bar{y}$ and $\sigma_z^2 = \sigma_x^2 + \sigma_y^2$

ii) In binary communications, bits 0 and 1 are transmitted as 000 and 111, respectively and majority rule is used for bit detection at the receiver. If $p_e = 10^{-5}$ be the error probability of one digit, then determine $P(e)$, the probability of making wrong decision.

b) i) How can Rayleigh density function be derived from two independent Gaussian RVs? Give an example in electrical communication system that follows Rayleigh density.

ii) Show that the variance of two independent random variables (RVs) is equal to the sum of their variances.

(3+2)+(3+2)

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech.(IT) 3rd Semester Final Examinations, 2023

Discrete Mathematics and Graph Theory (IT 2103)

Marks: 50

Time: 3 Hr

Answer any FIVE questions
Each question carries 10 marks

- ✓1. (a) Let $(G, *)$ be a group with subgroup H . Define a relation L on G so that aLb if and only if $b^{-1} * a \in H$. Prove that L is an equivalence relation.
(b) Show that the equivalence class containing $a \in G$ is the set $\{a * h \mid h \in H\}$.
(c) Is $(\mathbb{Z}_5, +)$ the quotient group of $(\mathbb{Z}, +)$ by the subgroup $5\mathbb{Z} = \{5x \mid x \in \mathbb{Z}\}$, where $\mathbb{Z}$ is the set of integers? Justify your answer.

3 + 3 + 4

- ✓2. (a) What is group morphism? If $f: G \rightarrow H$ is a group morphism, then prove that
(i) $f(e_G) = e_H$, where e_G and e_H are the identities of G and H respectively.
(ii) $f(a^{-1}) = f(a)^{-1}$ for all $a \in G$.
(b) What is group isomorphism? Prove that the group of non-zero real numbers under multiplication is isomorphic to the group of real numbers under addition.

5 + 5

- ✓3. (a) What is a lattice? Prove that a complemented, distributive lattice is a Boolean algebra.
(b) Suppose D_{48} denotes the set of all positive divisors of 48. Consider a relation $|$ on D_{48} , which is defined as follows: $x|y$ holds iff x divides y .
A. Test if the relation $|$ is a partial ordered relation. Draw the Hasse diagram for D_{48} .
B. Test if D_{48} is a lattice under $|$. If so, find out g.l.b and l.u.b.

5 + 5

4. (a) Is $(R, *)$ a semigroup or monoid or neither, where $x * y = \sqrt{x^2 + y^2}$? Justify your answer. Here, R is the set of real numbers and $x, y \in R$.
(b) Show that $(Q(\sqrt{2}), +, \cdot)$ is a commutative ring where $Q(\sqrt{2}) = \{a + b\sqrt{2} \in R \mid a, b \in Q\}$. Here, Q and R are the sets of rational and real numbers respectively.
(c) If a is a nonzero element of an integral domain $(R, +, \cdot)$ and $a \cdot b = a \cdot c$, then prove that $b = c$.

3 + 4 + 3

✓ 5. (a) Let $f: N \rightarrow R$ be a function, where N and R are the sets of natural and real numbers respectively. Prove that f cannot be surjective.

(b) State Correspondence Theory of Truth in Logic. State which of the following is Tautology, Contradiction and Contingent.

i. $\alpha \rightarrow (\beta \rightarrow \alpha)$

ii. $\alpha \wedge (\neg \beta)$

iii. $\alpha \wedge (\neg \alpha).$

5 + 5

✓ 6. (a) What is a permutation group? State the Cayley's theorem.

(b) Write short notes the following

(i) Hamiltonian cycle

(ii) Axiomatic set theory

(iii) Relational algebra

4 + 6

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech. (Information Technology) 3rd Semester Final Examination 2023
Digital Logic and Circuit Design (IT 2102)

Time-3 hours

Full Marks-50

Answer any five questions

- ✓✓ 1. a) Design a 4-bit minimized combinational logic using only NAND gates to produce an output 1 whenever the input is a multiple of 2 or 3.
b) Draw K-map for the following function of four variables and use them to reduce the function: $F(A, B, C, D) = \sum(0, 1, 4, 3, 5, 2, 11, 10)$
c) Verify the following by Boolean Algebra: $(A + \bar{B} + \bar{C})(A + \bar{B} + C) = A + \bar{B}$ (4+3+3)
2. a) Draw the circuit of a 4-to-1 multiplexer using SSI gates and explain the circuit operation.
b) Develop a logic circuit for a full adder with the help of a suitable decoder and other gates. (Use the block diagram representation for the decoder.)
c) Derive 3-bit gray code by reflecting 2-bit gray code? Design a 4-bit Binary to Gray Code converter. (3+3+4)
- ✓✓ 3. a) Convert the following function into the circuit using the NOR/NAND universal logic:
 $F = (A+B+C)(A+B'+C')$
b) Simplify the expression $f = (A+B).(A'+C)$ and realize it using NOR gate only.
c) Find the essential prime implicant, non-essential prime implicant of the following function, $F = \sum(0, 1, 3, 4, 5, 7, 8, 9, 13, 15)$ and also write the SOP expression. (3+3+4)
4. a) Draw the logic diagram of a 2-to-4 line decoder with only NOR gates include an enable input.
b) Construct a 5-to-32-line decoder with 3-to-8 line decoders with enable and one 2-to-4 line decoder.
c) Implement the function $F(A, B, C, D) = \sum(1, 3, 4, 11, 12, 13, 14, 15)$ using 8x1 MUX. (3+3+4)
- ✓✓ 5. a) What is the race-around problem in JK flip-flop? How can it be avoided?
b) The following pulse trains are arriving at J & K inputs of a J-K F/F simultaneously. J = 100001, K = 000101, Draw the truth table of the operation.
c) Realize a T-flip flop using one JK-flip flop and some basic logic gates (if needed). (3+3+4)
- ✓✓ 6. a) Why is asynchronous counter called Ripple Counter? How many F/Fs are required to design a MOD-20 counter?
b) Explain the design and operation of a 4-bit down ripple Counter.
c) Design a synchronous counter to produce a sequence given in decimal form 0, 2, 1, 3, 6. (3+3+4)
- ✓✓ 7. a) What is buffer register? What are the applications of register?
b) Draw and explain the operation of a 4-Bit right shift register circuit using JK flip-flops.
c) With the help of necessary circuit diagram, explain the operation of a 4-bit Johnson counter. (3+4+3)
8. Write short technical notes on (Any four)
(a) Realization of a 2-input XOR gate using minimum number of 2-input NAND gates
(b) Use of NOR as universal logic gate
(c) Internal circuit of J-K master slave flip-flop
(d) De MORGAN's theorem and its' proof
(e) 2's Complement Adder/Subtractor
(f) Gray code is a cyclic code-Explain (2.5x4)

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech 3rd Semester End-Semester Examination, November 2023

Mathematics-III(MA 2101)

Full Marks : 50

Time : 3 hours

Answer any five questions

1. a) The probability that at least one of the events A and B occurs is $\frac{3}{5}$. If A and B occur simultaneously with probability $\frac{1}{5}$, then find $P(\bar{A}) + P(\bar{B})$, where $\bar{A}$ and $\bar{B}$ are the complementary events of A and B respectively.

b) Given for any two events A and B , $P(A) = \frac{1}{4}$, $P(B/A) = \frac{1}{2}$ and $P(A/B) = \frac{1}{4}$; show that A and B are independent.

c) The probability that a man live next 25 years is $\frac{3}{5}$ and that of his wife is $\frac{2}{3}$. Find the probability of that

(i) both will alive (ii) only the man be alive (iii) only the wife will be alive (iv) at least one of them will be alive.

d) Examine whether the function $F(x) = \frac{2}{\pi} \tan^{-1} x$, $(-\infty < x < \infty)$ is a distribution function.

[2+2+4+2]

2. a) State and prove Tchebycheff inequality for a continuous random variable.

b) The probability density function of a two-dimensional random variable (X, Y) is given by

$$f(x, y) = \begin{cases} k(x+y), & x \geq 0, y \geq 0, 0 < x+y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find k and evaluate $P(X < \frac{1}{2}, Y > \frac{1}{4})$.

[(1+5)+(2+2)]

3. a) Show that the correlation coefficient (ρ) between two random variables X and Y , satisfies the relation: $\rho^2 \leq 1$.

b) Fit a straight line to the following data:

x_i	y_i
1	2.4
2	3
3	3.6
4	4
6	5
8	6

[5+5]

4. a) Define Laplace transform of a function $f(t)$, ($t \geq 0$).
 b) Find $L\{e^{-3t}(2\cos 5t - 3\sin 5t)\}$.
 c) Solve the following differential equation by Laplace transform:

$$(D^2 + 6D + 9)y = 0; y(0) = y'(0) = 1$$

[1+4+5]

5. a) Let X is Poisson variate with $P(1) = P(2)$. Find (i) mean and (ii) $P(X = 4)$.
 b) If the mean of a Binomial distribution is 3 and the variance is $\frac{3}{2}$, find the probability of obtaining at most 3 successes.
 c) If X follows a normal distribution with mean 12 and variance 16, then find the value of $P(X \geq 20)$. [Given: $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^2 e^{-t^2/2} dt = 0.977725$]

[(1+1)+4+4]

6. a) Show that $X = \{(x_1, x_2, x_3) \in E^3; 2x_1 + 3x_2 + x_3 \leq 5\}$ is a convex set.
 b) Use simplex method to solve the following L.P.P :

$$\begin{aligned} \text{Max } z &= x_1 + x_2 + 3x_3 \\ \text{Subject to: } 3x_1 + 2x_2 + x_3 &\leq 3 \\ 2x_1 + x_2 + 2x_3 &\leq 2 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

[3+ 7]

7. a) Find all the basic solutions to the system of linear equations

$$\begin{aligned} x_1 + x_2 + x_3 &= 4 \\ 2x_1 + 5x_2 - 2x_3 &= 3. \end{aligned}$$

- b) $x_1 = 2, x_2 = 3, x_3 = 1$ is a feasible solution of the system of equations:

$$\begin{aligned} 2x_1 + x_2 + 4x_3 &= 11 \\ 3x_1 + x_2 + 5x_3 &= 14. \end{aligned}$$

[4+6]

Reduce this feasible solution to a basic feasible solution.

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR
B. Tech.(IT) 3rd Semester Final Examination, 2023

Data Structure and Algorithms (IT-2101)

Full Marks: 50

Time: 3 Hours

Answer any five questions

- ✓ 1. Explain with example the difference between definiteness and effectiveness property of an algorithm. When is an algorithm considered as correct? What do you mean by asymptotic notation? Show that $10n^2 + 5n$ is $\Theta(n^2)$ and $n^2 + 10n + 5$ is $\Omega(n^2)$. Why is array considered as linear data structure? Consider that an n -dimensional array of size $m_1 \times m_2 \times \dots \times m_n$ is stored in row-major form. If the base address of the array is BA , then calculate the address of a random element present in this array. 2+1+1+3+1+2
2. An $n \times n$ square matrix A is said to be symmetric if $A[j, k] = A[k, j]$ for all j and k . Now suppose A and B are two $n \times n$ square symmetric matrices. Describe an efficient way of storing A and B in memory. Consider the expression written in postfix form: $8\ 6\ 2\ -\ /\ 2\ 3\ 5\ +\ * +$. Find out the value of the above expression using stack. Can we represent double ended queue as a circular array? If yes, then write down the conditions for *queue full* and *queue empty*. Explain the advantages and disadvantages of linked list over array. 2+2+3+3
3. Consider a double linked list L containing integer values. Write an algorithm to rearrange the double linked list L in such a way that all the nodes less than an integer x comes before all the nodes greater than or equal to x . Suppose linked list is used to represent polynomial. Write an algorithm to multiply the two polynomials P and Q , and store the result in another polynomial R . Write an algorithm to reverse the elements of a single circular linked list. 3+4+3
- ✓ 4. Assume that a priority queue is represented by a linked list where each node contains data as well as priority value. If two nodes have same priority then the node inserted first will be deleted first. Write an algorithm to insert an element with data K and priority P in the priority queue. Prove that in a binary tree the maximum number of nodes on level i ($i \geq 0$) is 2^i . Explain the problems of using array for representing binary tree. Draw the expression tree for the expression $(5a + b)(6c - 4)^3$. 4+2+2+2

5. Draw a binary search tree using the data sequence 70, 80, 30, 50, 80, 20, 75, 45. Now delete the node with data 30 and show the final tree. Explain the insertion of a node in a threaded binary tree. What is a heap? What do you mean by height balanced tree? Draw an AVL (height balanced tree) using the following data sequence: 50, 30, 60, 55, 25, 35, 15. Now delete the nodes in the sequence 55, 35. Then insert the nodes in the sequence 26, 28. Finally delete the node 60. Show the trees after each insertion and deletion. Also mention the type of rotation(s) if any.

2+3+1+1+3

6. Explain the insertion and deletion operations in a B-tree. Explain the scenario when you prefer to use linear search over binary search. Write the non-recursive algorithm for binary search method. What are the properties of a good hash function? Explain some popular hash functions with example.

4+1+2+1+2

7. Write the algorithm for sorting a set of numbers using selection sort method and also calculate the worst time complexity of the algorithm. What is in-place sorting? Write the algorithm for sorting a set of numbers using merge sort method and also calculate the worst time complexity of the algorithm.

4+1+5